Geoffrey Bian

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Education

University of British Columbia — September 2022 – May 2027
Bachelor of Applied Science, Computer Engineering, GPA 4.0/4.0 — *Vancouver, BC*

Technical Skills

Languages: Python, C++, C, Rust, JavaScript (React), Java, Go, SQL
Systems & Tools: Claude Code, Docker, MATLAB (Simulink), Linux, Git, AWS EC2, SPI, CAN, RTOS
Domains: Control/Embedded Systems, Firmware, Robotics, Computer Networking, Machine Learning

Work Experience

Tesla — May 2026 – Present
Software Engineer Intern – Crash Safety — *Palo Alto, CA*

- Building **Hardware-in-the-Loop** (HIL) test infrastructure to enable automated validation across the **Restraint Control Module** driver stacks, peripheral interfaces, and vehicle communication pathways.
- Leveraging **AI-assisted engineering workflows** by developing custom Claude skills and autonomous testing agents to accelerate test generation, failure triage, and validation coverage across embedded safety systems.

Tesla Energy — February 2025 – August 2025
Software Engineer Intern – SCADA — *Palo Alto, CA*

- Designed scalable **automation and control logic** using *Python*, *Rust*, and *Structured Text*, integrating telemetry and fault detection to enhance system uptime and data reliability across 12+ **Balance of Plant** subsystems.
- Developed predictive **Battery Energy Storage System** and **Solar** supervisory control algorithms in *MATLAB Simulink*, optimizing real/reactive power dispatch for deployment on 40MWh+ industrial sites.
- Recognized with a top-tier performance evaluation for exceptional technical execution, ownership, and impact.

Rivian and Volkswagen Group Technologies — May 2024 – February 2025
Software Engineer Intern – Body Controls (with Extension) — *Irvine, CA and Vancouver, BC*

- Developed customer-facing **Body Control** features (**Vehicle Access** and **Exterior Lighting**) in *C* and on an *RTOS* environment to adhere to system requirements, *MISRA* coding standards, and rigorous *ASIL A/B/C* procedures.
- Integrated 30+ **key performance indicators** to capture user-vehicle interaction data, upholding functional safety standards and supporting data-driven decision-making.
- Performed in-vehicle **system integration** and leveraged *CANape* (*Vector*) tools to analyze CAN, LIN, and Ethernet signals, perform root cause analysis, and resolve 20+ firmware-to-system inconsistencies.
- Optimized build and dependency management with *Bazel*, improving scalability of large-scale system architecture.

Acuren — May 2023 – August 2023
Mechanical Engineering Intern — *Richmond, BC*

- Analyzed large-scale **Stress-Strain Datasets** (15,000+ data points) to predict material failure using regression and curve-fitting techniques in *Ansys* and *Excel*.

Academic Experience

UBC Baja SAE — August 2024 – Present
Software Lead — *Vancouver, BC*

- Providing **technical leadership and mentorship** to a sub-team of 9 members, guiding them through embedded development, CAN debugging, and mechatronic systems integration.
- Building a **mechatronics system** for IMU, tachometer, speedometer, and suspension sensing and control with STM32 and Raspberry PI ECUs running algorithms communicating over distributed CAN, SPI, and I2C interfaces.

UBC CIRRUS Lab — September 2025 – Present
Undergraduate Research Assistant — *Vancouver, BC*

- Co-authored GreenPASS (to appear at ACM e-Energy 2026) developing a **region-aware optimization** for workload shifting, modeling energy usage, latency, and transfer overhead to minimize cost and **carbon emissions**.
- Designed analytical models for power, carbon intensity (CI), and shifting overhead (network + storage), and integrated forecasting methods (LightGBM, SES) to enable robust scheduling under uncertain grid conditions.